

has become necessary to validate if the application is supported on other browser versions without compatibility issues. If the application is not behaving properly on certain browsers, it is good to mention the supported versions to avoid customer complaints. Below are some of the common checkpoints:

- Checks browser rendering of your application's user interface.
 - Checks the browser's security settings for cross-domain access and hacks.
 - Verifies consistent functioning of the app across multiple versions of a browser.
 - Checks user interface rendering on different-sized mobile device screens, including screen rotation.
 - Verifies that the application operates correctly when the device moves in and out of the range of network services.
3. **Performance testing:** Performance testing focuses on checking how an application behaves under extra load, which refers to the number of users accessing the application simultaneously. It is good to see which particular feature is breaking down under the given load. Listed below are some of the common checkpoints:
- Checks the server's response to the browser form submit requests.
 - Identifies changes in performance over a period of time.
 - Tests for functions that stop working at higher loads.
 - Identifies how an application functions after a system crash or component failure.
 - Identifies forms and links that operate differently under higher loads.
4. **Security testing:** Securing user data is a critical task and Web apps should not leak data. Testing ensures that the app works only with a valid login, and that after logout, the data remains secure and pressing the 'back' key does not resume the session. Given below are some of the common checkpoints:
- Checks whether the app operates on certain URLs without logging.
 - Tests basic authentication using false user names and password credentials.
 - Checks if the app functions correctly upon invalid URL attribute values.
 - Checks how the app functions with invalid input fields, including text fields.
 - Tests CAPTCHA fields for Web forms and logins.
5. **Usability testing:** Any Web app is considered user friendly if accessibility is easy and navigation is smooth. If there are ambiguities in representations, then these should be corrected. Users want clear descriptions and representations. Shown below are some of the common checkpoints:
- Tests that the content is logically arranged and easy for users to understand.
 - Checks for spelling errors.
 - Checks that pages adhere to colour and pattern style

guidelines, including fonts, frames and borders.

- Checks that images load correctly and in their proper size.
- With the increasing need to analyse the performance of your Web app, it is a good idea to evaluate some of the popular open source performance testing tools.

Why choose open source performance test tools?

1. No licensing costs – a commercial load testing tool can really burn a hole in your pocket when you want to test with a large number of virtual users.
2. Generates (almost) an infinite amount of load on the Web app without charging users any additional licensing costs. The only limitation would be the resources available.
3. Enables you to create your own plugins to extend the analysis and reporting capabilities.
4. Integrates with other open source and commercial tools to drive end-to-end test cycles.

Popular open source Web application test tools

Licensed tools have their own benefits but open source always stands out because of the ease of use. Here are some popular open source Web app test tools that are easily available and simple to use as well.

1. JMeter: Load and performance tester

JMeter is a pure Java desktop application designed to load-test functional behaviour and measure performance. It can be used to test performance both on static and dynamic resources (files, Servlets, Perl scripts, Java objects, databases and queries, FTP servers and more). It can be used to simulate a heavy load on a server, network or object to test its strength or to analyse the overall performance under different load types. JMeter was originally used for testing Web and FTP applications. Nowadays, it is used for functional tests, database server tests, etc.

The pros of JMeter

- A very lightweight tool that can be installed easily.
- As it is an open source tool, you need not be worried about the licence.
- There are multiple plugins that are available in the market



Figure 1: JMeter